of an activity are known as tail event and head event respectively. Event is generally represented by a numbered circle. The head event, called the jth event, has always a number higher than the tail event, called the ith event, ie., $j > i$.

Predecessor Activity:

Activities which must be completed before a particular activity starts are called predecessor activities. If an activity A is predecessor of an activity B, it is denoted by $A<B$. (i.e.,) activity B can start only if activity A is completed.

Successor Activity:

An activity that cannot be started until one or more of the other activities are completed, but immediately succeed them is referred to as successor activity.

Network:

Network is a diagrammatic representation of various activities concerning a project arranged in a logical manner.

Path:

A path is defined as a set of nodes connected by lines which begin at the initial node and end at the terminal node of the network.

10.2.1 Construction of network:

Rules for constructing network

For the construction of a network, generally, the following rules are followed:

(i) Each activity is represented by one and only one arrow. (i.e) only one activity can connect any two nodes.

(ii) No two activities can be identified by the same head and tail events.

(iii) Nodes are numbered to identify an activity uniquely. Tail node (starting point) should be lower than the head node (end point) of an activity.

(iv) Arrows should not cross each other.

(v) Arrows should be kept straight and not curved or bent.

(vi) Every node must have atleast one activity preceding it and atleast one activity following it except for the node at the beginning and at the end of the network.

Numbering the Events

After the network is drawn in a logical sequence, every event is assigned a number. The number sequence must be such as to reflect the flow of the network. In event numbering, the following rules should be observed:

(i) Event numbers should be unique.

(ii) Event numbering should be carried out on a sequential basis from left to right.

(iii) The initial event is numbered 0 or 1.

(iv) The head of an arrow should always bear a number higher than the one assigned at the tail of the arrow.

(v) Gap should be left in the sequence of event numbering to accommodate subsequent inclusion of activities, if necessary.

Remark: The above procedure of assigning numbers to various events of a network is known as Fulkerson's Rule.

Example 10.9

Draw the logic network for the following:

Activities C and D both follow A, activity E follows C, activity F follows D, activity E and F precedes B.

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